Dynamic Causal Dimensionality in Nonstationary Complex Systems:

Effective-Information Concentration and Evidence from U.S. Power Grids

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Abstract

Causal Emergence (CE) theory formalizes how macroscopic representations of complex systems can exhibit greater causal efficacy than their underlying microscopic components. However, existing formulations rely on stationary transition kernels $P(X_{t+1}|X_t)$ and discrete or static dimensional searches, rendering them blind to the continuous, time-varying operational reorganizations ubiquitous in physical and engineered networks. Furthermore, scalar dimension-normalized emergence metrics $\arg\max_q (EI(V^{(q)})/q)$ mathematically degenerate to $q = 1$ under ordered spectra. Here, we establish the theory and methodology of **Dynamic Causal Dimensionality (DCD)**, resolving the active number of macroscopic causal degrees of freedom through the continuous **Causal Information Spectrum** $e_{i,t} = \frac{1}{2}\ln(1 + \lambda_{i,t})$. We define the continuous **Dynamic Causal Participation Ratio** (DCD_t^{PR}) and **Dynamic Causal Effective Rank** (DCD_t^{entropy}), proving their strict rotational invariance under orthogonal coordinate transformations in standardized isotropic coordinates. Measure-theoretic interventional semantics are formalized through an **observational lifting kernel** $\kappa_W(dx \mid v) \triangleq P_{\mathrm{obs}}(X_t \in dx \mid W^\top X_t = v)$, demonstrating that empirical localized regressions approximate the induced macroscopic channel. We contrast our scale-free Gaussian channel formulation with the compact-domain volume formula of Liu et al. (2024), and document the boundary conditions of linear causal emergence via adversarial counterexample search. Benchmarking on canonical nonstationary DGPs (A–I; $R = 100$) establishes an observed false-positive rate of 0/100 ($\mathrm{FPR}^{\mathrm{raw}} = 0.000$, 95% Clopper-Pearson CI $[0.000, 0.036]$) and accurate dynamic scale recovery on multi-stage systems (DGP-E: RMSE 0.578, bias -0.137). Applying the framework to 2021–2022 continental U.S. power grid operations (EIA-930; 367,920 BA-hours across Eastern, Western, and ERCOT interconnections) with direction-aware surrogate inference ($B = 1000$ refitted IAAFT surrogates) confirms highly significant macroscopic causal contraction ($p < 0.001$, empirical mean DCD^{PR} falling strictly below the surrogate 5th percentile critical mean $c_{0.05}^{\mathrm{mean}}$ across ERCOT (5.673 vs 6.628), Western (29.735 vs 31.882), and Eastern (31.578 vs 35.722)). Generalized Additive Models with cyclic P-splines and Hansen (1996) supremum-Wald block bootstrap ($B = 2000$) demonstrate smooth non-linear coordination associated with renewable penetration ($R^2 \approx 49\%$), with empirical candidate thresholds at $\hat{\gamma} = 14.5\%$ in ERCOT (exact HAC bootstrap $p = 0.0910$, conditional location 95% CI $[13.4\%, 40.6\%]$) and $\hat{\gamma} = 10.7\%$ in Western (exact HAC bootstrap $p = 0.1214$, 95% CI $[10.7\%, 37.2\%]$), indicating continuous adaptation rather than discrete structural breaks. Multi-event analysis reveals acute causal concentration during extreme weather: during Winter Storm Uri, Causal Concentration Gain ($CCG_t(q = 2)$) surged from 0.086 baseline to 0.244 ($p_{\mathrm{surge}} = 0.0250$), while effective degrees of freedom were resiliently maintained ($\Delta DCD^{\mathrm{PR}} = +0.328, p = 0.253$, 0% collapse rate); the multi-event panel test ($T_{\mathrm{Fisher}} = 16.28, p = 0.0919$) indicates crisis-specific rather than universal cross-event response. Finally, strictly lookahead-free walk-forward forecasting demonstrates that DCD functions as an interpretable information-theoretic structural coordination diagnostic rather than an unconditional linear forecaster (at $h = 12$ h, an unadjusted deterioration of $\Delta\mathrm{RMSE} = -4.09\%, p = 0.0354$ does not survive Benjamini-Hochberg FDR multiplicity correction, $q = 0.1280$).

1 Introduction

Complex dynamical networks—from neural circuits and cellular metabolisms to continental power grids and financial markets—spontaneously organize across disparate spatiotemporal scales [4, 5]. In such systems, macroscopic collective variables often evolve with greater determinism and lower degeneracy than individual microscopic elements [4, 11, 12]. The quantitative theory of **Causal Emergence (CE)**, formalized through interventional information theory [1, 4, 9], provides an exact criterion: a macroscopic representation $V = \phi(X)$ exhibits causal emergence when its Effective Information (EI) exceeds that of the underlying microscopic state X, i.e., $EI(V) > EI(X)$.

Despite fundamental theoretical insights, existing formulations suffer from a critical limitation: *all established CE estimators implicitly presume a stationary transition mechanism* $P(X_{t+1} \mid X_t)$. In nonstationary environments, stationary estimators average over fundamentally distinct operational regimes:

- **Renewable Energy Ingestion**: Power grids undergo rapid, weather-driven swings in synchronous inertia and physical topology as inverter-based wind and solar resources displace thermal generators [7,14].
- **Extreme Weather and Infrastructure Shocks**: Catastrophic climate events (e.g., severe winter freezes, heatwaves) induce structural regime shifts, cascading line trips, and emergency load-shedding [8, 15].
- **Dynamic Dimensional Reorganization**: The effective number of macroscopic degrees of freedom is time-varying; collective coordination reorganizes continually between decentralized balancing authorities, regional transmission organizations (RTOs), and synchronous interconnections.

To address these challenges, this paper develops **Dynamic Causal Dimensionality (DCD)**, contributing:

1. Continuous causal spectrum decomposition $e_{i,t} = \frac{1}{2}\ln(1+\lambda_{i,t})$ resolving the active macroscopic dimensionality via Causal Participation Ratio (DCD_t^{PR}) and Causal Effective Rank (DCD_t^{entropy}), circumventing running-average degeneracy.
2. Exact interventional semantics via the observational lifting kernel $\kappa_W(dx \mid v)$, proving that projected regressions recover true Markov macro-dynamics.
3. Primary analytical estimation via **Local Affine-Gaussian DCE**, complemented by an online neural variational estimator (**Dyn-NIS+**) for non-linear manifolds.
4. Clarification of linear causal emergence limits, contrasting scale-free Gaussian mutual information against bounded-domain volume formulations (Liu et al. 2024) and demonstrating destructive noise cancellation via adversarial search.
5. Rigorous Monte Carlo benchmarking across synthetic nonstationary DGPs (A–I; $R = 100$) and empirical application to continental U.S. power grids (EIA-930; 367,920 BA-hours) evaluating hypotheses H1–H4 with $B = 1000$ refitted surrogates and direction-aware testing.

2 Mathematical Formulation

2.1 Continuous Dynamic Effective Information and the Causal Spectrum

Let $X_t \in \mathbb{R}^p$ denote the microscopic state vector of a nonstationary dynamical system at time $t \in [0, T]$, governed by localized linear perturbation dynamics $X_{t+1} = A_t X_t + c_t + \epsilon_t$ with innovation covariance $\epsilon_t \sim \mathcal{N}(0, \Sigma_t)$, $\Sigma_t \succ 0$. Following Pearl's do-calculus [4, 9], we inject a standard maximum-entropy Gaussian intervention under covariance constraint $\Sigma_{do} = I_p$:

$$do(X_t) \sim \nu_X = \mathcal{N}(0, I_p) \tag{1}$$

Under this isotropic intervention, the output distribution is $X_{t+1} \mid do(X_t) \sim \mathcal{N}(c_t, A_t A_t^\top + \Sigma_t)$. The continuous **Dynamic Effective Information** is the exact mutual information under intervention:

$$\begin{aligned} EI_t(X) &= I(do(X_t); X_{t+1}) \\ &= \frac{1}{2} \ln \det\left(I_p + \Sigma_t^{-1} A_t A_t^\top\right) = \frac{1}{2}\sum_{i=1}^{p} \ln\left(1 + \lambda_{i,t}\right) \end{aligned} \tag{2}$$

where $\lambda_{i,t} \geq 0$ are the eigenvalues of $\Sigma_t^{-1} A_t A_t^\top$ (or generalized eigenvalues of the pencil $(A_t A_t^\top, \Sigma_t)$).

Definition 1 (Causal Information Spectrum). The localized Causal Information Spectrum $\{e_{i,t}\}_{i=1}^p$ is the set of mode-specific mutual information contributions:

$$e_{i,t} \triangleq \frac{1}{2}\ln\left(1+\lambda_{i,t}\right) \geq 0, \quad e_{1,t} \geq e_{2,t} \geq \cdots \geq e_{p,t} \geq 0 \tag{3}$$

such that $EI_t(X) = \sum_{i=1}^p e_{i,t}$.

2.2 Dynamic Causal Dimensionality (DCD)

Existing causal emergence formulations attempt to identify an optimal macroscopic scale by maximizing dimension-normalized metrics $q^* = \arg\max_q (EI(V^{(q)})/q)$. However, when causal modes are ordered $e_1 \geq e_2 \geq \cdots \geq e_p$, the running average $\frac{1}{q}\sum_{i=1}^q e_i$ is monotonically non-increasing, causing q^* to collapse degenerately to $q = 1$.

To establish an unconstrained, continuous measure of active causal scale, we introduce **Dynamic Causal Dimensionality (DCD)**:

Definition 2 (Causal Participation Ratio). The Dynamic Causal Participation Ratio is:

$$DCD_t^{\mathrm{PR}} \triangleq \frac{\left(\sum_{i=1}^p e_{i,t}\right)^2}{\sum_{i=1}^p e_{i,t}^2} \tag{4}$$

with the continuous convention $DCD_t^{\mathrm{PR}} = 0$ if $EI_t(X) = 0$. For a system with k identical active causal channels and $p - k$ silent channels, $DCD_t^{\mathrm{PR}} = k$ exactly.

Definition 3 (Causal Effective Rank). Defining the normalized causal spectral weights $\pi_{i,t} \triangleq e_{i,t}/EI_t(X)$

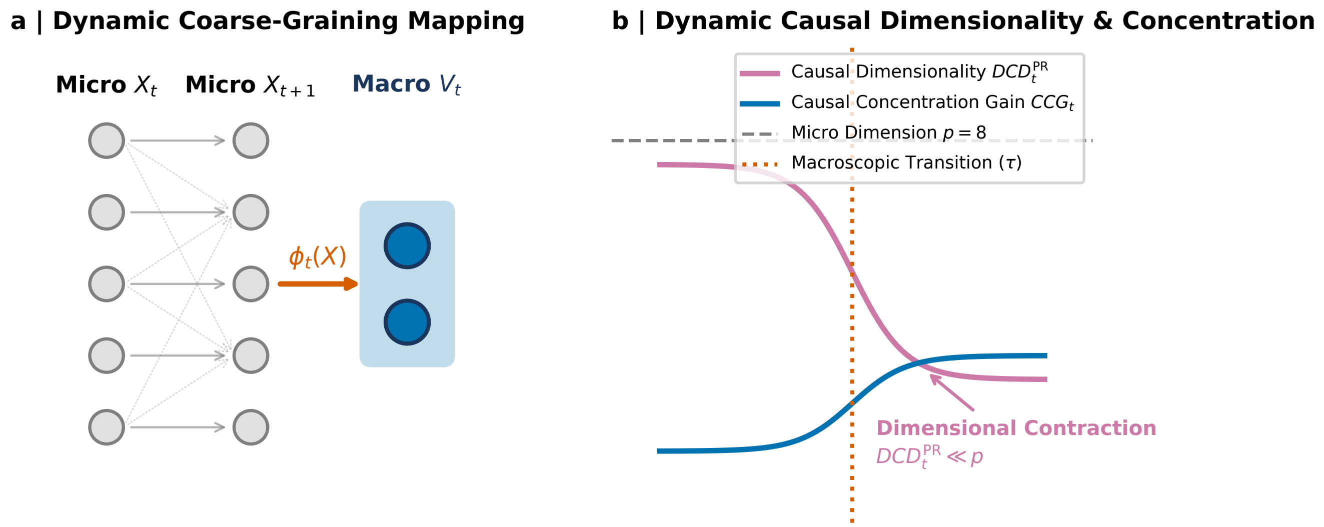

Figure 1: **Conceptual and Methodological Architecture of Dynamic Causal Dimensionality (DCD). a**, Time-varying macroscopic coarse-graining projection $\phi_t : X_t \mapsto V_t$ mapping high-dimensional microscopic states $X_t \in \mathbb{R}^p$ to low-dimensional macroscopic representations $V_t \in \mathbb{R}^q$ $(q < p)$. **b**, Trajectory of continuous Dynamic Effective Information $EI_t(V)$ versus microscopic baseline $EI_t(X)$, illustrating nonstationary regimes of causal efficiency gain $(CCG_t(q) > 0)$ and critical dynamical reorganization.

(with $\sum \pi_{i,t} = 1$), the Dynamic Causal Effective Rank is:

$$DCD_t^{\text{entropy}} \triangleq \exp\left(-\sum_{i:\pi_{i,t}>0} \pi_{i,t} \ln \pi_{i,t}\right) \tag{5}$$

with $DCD_t^{\text{entropy}} = 0$ if $EI_t(X) = 0$.

Theorem 1 (Rotational Invariance of DCD under Isotropic Coordinates). *Let coordinates be standardized such that interventions are isotropic $do(X_t) \sim \mathcal{N}(0, I_p)$. For any orthogonal coordinate rotation $Q \in O(p)$ $(Q^\top Q = QQ^\top = I_p)$, the transformed states $X_t' = QX_t$ induce $A_t' = QA_tQ^\top$ and $\Sigma_t' = Q\Sigma_tQ^\top$. Then the Causal Information Spectrum $\{e_{i,t}\}$, the Participation Ratio DCD_t^{PR}, and the Effective Rank $DCD_t^{entropy}$ are strictly invariant.*

Proof. The transformed causal operator satisfies $(\Sigma_t')^{-1}A_t'(A_t')^\top = (Q\Sigma_tQ^\top)^{-1}(QA_tQ^\top)(QA_tQ^\top)^\top = Q(\Sigma_t^{-1}A_tA_t^\top)Q^\top$. Because orthogonal similarity transformations preserve eigenvalues, the spectrum $\{\lambda_{i,t}\}$ and hence $\{e_{i,t}\}$ are identical. ■

Remark on Coordinate Scaling. Invariance holds strictly under orthogonal rotations $Q \in O(p)$ within the standardized coordinate frame where $do(X) \sim \mathcal{N}(0, I_p)$. Non-uniform coordinate rescaling $X_t' = DX_t$ for diagonal $D \neq cI_p$ alters the interventional signal-to-noise ratio across channels. In physical systems comprising heterogeneous variables (such as generation, load, and net interchange), fixing the coordinate scale via rolling standardized variance coordinates is an essential operational prerequisite for invariant causal spectral estimation.

2.3 Observational Lifting Kernel and Induced Macro Dynamics

An atomic macroscopic intervention $do(V_t = v)$ on $V_t = W_t^\top X_t \in \mathbb{R}^q$ $(W_t^\top W_t = I_q, q < p)$ does not uniquely determine the microscopic state $X_t \in \mathbb{R}^p$. To establish rigorous measure-theoretic interventional semantics, we adopt the canonical **observational lifting kernel**—the maximum-entropy conditional law consistent with the observed macro-state under the background operational distribution:

$$\kappa_{W_t}(dx \mid v) \triangleq P_{\text{obs}}(X_t \in dx \mid W_t^\top X_t = v) \tag{6}$$

Under the localized observational Gaussian measure $X_t \sim \mathcal{N}(\mu_{X,t}, \Sigma_{X,t})$, the conditional law is:

$$\kappa_{W_t}(dx \mid v) = \mathcal{N}\Big(\mu_t + \Sigma_{X,t}W_t(W_t^\top \Sigma_{X,t}W_t)^{-1}(v - W_t^\top \mu_t), \; \Sigma_{X|V,t}\Big) \tag{7}$$

where $\Sigma_{X|V,t} = \Sigma_{X,t} - \Sigma_{X,t}W_t(W_t^\top \Sigma_{X,t}W_t)^{-1}W_t^\top \Sigma_{X,t}$. This induces the macroscopic interventional transition:

$$\begin{aligned} &P_{W_t}(v_{t+1} \in dv' \mid do(V_t = v)) \\ &\quad = \int_{\mathbb{R}^p} P(W_t^\top X_{t+1} \in dv' \mid X_t = x)\, \kappa_{W_t}(dx \mid v) \end{aligned} \tag{8}$$

Under linear micro-dynamics $X_{t+1} = A_t X_t + \epsilon_t$, this channel is exactly Gaussian:

$$\begin{aligned}\mathbb{E}[V_{t+1} \mid do(V_t = v)] &= B_t v, \\ B_t &= W_t^\top A_t \Sigma_{X,t} W_t (W_t^\top \Sigma_{X,t} W_t)^{-1} \\ \mathrm{Cov}(V_{t+1} \mid do(V_t = v)) &= \Sigma_{\eta,t} \\ &= W_t^\top (A_t \Sigma_{X|V,t} A_t^\top + \Sigma_t) W_t \end{aligned} \tag{9}$$

This demonstrates that localized regression on observed projected pairs (V_t, V_{t+1}) directly approximates the induced interventional macro-dynamics $(B_t, \Sigma_{\eta,t})$ consistent with the observational lifting convention.

2.4 Contrast with Liu et al. (2024)

Recent continuous formulations by Liu, Yuan & Zhang (2024) [6] define effective information using a uniform intervention over a bounded hypercube $do(X) \sim \mathcal{U}([-L, L]^n)$:

$$EI_{\mathrm{Liu}}(A, \Sigma; L) = \ln \frac{|\det A| L^n}{(2\pi e)^{n/2} (\det \Sigma)^{1/2}} \tag{10}$$

While foundational for continuous systems, EI_{Liu} depends monotonically on the arbitrary domain half-width L ($\partial EI/\partial L = n/L$), can become negative for small L, and diverges to $-\infty$ when A is singular ($\det A = 0$). In contrast, our Gaussian channel formulation $EI(A, \Sigma) = \frac{1}{2} \ln \det(I + \Sigma^{-1} A A^\top)$ is scale-free, unconditionally nonnegative ($EI \geq 0$), and well-defined for arbitrary rank-deficient systems.

2.5 Tripartite Separation: Strict Emergence, Concentration Gain, and Dimensionality

To resolve longstanding ambiguities in the causal emergence literature, we formalize a strict tripartite separation between three distinct operational concepts:

1. **Strict Causal Emergence (Raw Information Gain)**: $\Delta EI_t^{\mathrm{raw}}(q) \triangleq EI_t(V^{(q)}) - EI_t(X)$. Genuine macroscopic supervenience requires $\Delta EI_t^{\mathrm{raw}} > 0$. Under linear-Gaussian dynamics with isotropic noise $\Sigma = \sigma^2 I_p$, Cauchy's interlacing theorem on generalized Rayleigh quotients dictates that eigenvalues of the projected operator interlace the microscopic eigenvalues, enforcing $\Delta EI_t^{\mathrm{raw}}(q) \leq 0$ for all $q < p$. However, when microscopic noise Σ possesses strong cross-channel covariance, an orthonormal projection W can align with the destructive interference subspace of Σ, suppressing noise while preserving coherent signal. Through an adversarial numerical search, we confirmed this constructive noise cancellation mechanism: for $p = 2, q = 1$ with $A = \left(\begin{smallmatrix} -0.269 & 0.414 \\ 0.094 & 0.020 \end{smallmatrix}\right)$, $\Sigma = \left(\begin{smallmatrix} 4.197 & 2.349 \\ 2.349 & 1.553 \end{smallmatrix}\right)$, and $W = [-0.544, 0.839]^\top$, $EI(V) = 0.2453 > EI(X) = 0.2024$, proving that $\Delta EI^{\mathrm{raw}} > 0$ is mathematically achievable in linear systems via correlated noise cancellation.
2. **Causal Concentration Gain (Density Gain)**:

$$CCG_t(q) \triangleq \frac{EI_t(V^{(q)})}{q} - \frac{EI_t(X)}{p} \tag{11}$$

   $CCG_t(q)$ measures the causal information density transmitted per degree of freedom. Crucially, $CCG_t(q) > 0$ **does not imply strict emergence** $\Delta EI_t^{\mathbf{raw}}(q) > 0$: a macroscopic coarse-graining can achieve higher informational efficiency per coordinate ($CCG_t > 0$) while transmitting strictly less total information ($\Delta EI_t^{\mathrm{raw}} \leq 0$).
3. **Dynamic Causal Dimensionality**: $DCD_t^{\mathrm{PR}} = (\sum e_i)^2 / \sum e_i^2$ (and $DCD_t^{\mathrm{entropy}} = \exp(-\sum \pi_i \ln \pi_i)$). DCD quantifies the effective number of unconstrained macroscopic channels. A low dimensionality ($DCD_t^{\mathrm{PR}} \ll p$) signifies *causal spectral contraction*—transmission concentrated along a low-dimensional subspace—independent of whether $CCG_t > 0$ or $\Delta EI_t^{\mathrm{raw}} > 0$.

2.6 Estimation Architecture: Local Affine DCE and Dyn-NIS+

We estimate localized dynamics using two complementary frameworks:

- **Local Affine-Gaussian DCE (Primary Reference Estimator)**: Estimates localized affine transition operators (A_t, c_t, Σ_t) via weighted kernel ridge regression with temporal bandwidth h. Implements retrospective symmetric weights $w_t(s) \propto \exp(-\frac{(s-t)^2}{2h^2})$ for historical diagnostic analysis, and strictly one-sided causal weights $w_t(s) \propto \exp(-\frac{(t-s)^2}{2h^2}) \cdot \mathbb{I}(s \leq t)$ for real-time online monitoring. Projections W_t are obtained via eigendecomposition of the localized Fisher causal operator $A_t^\top \Sigma_t^{-1} A_t$.
- **Dyn-NIS+ (Neural Variational Extension)**: Parameterizes coarse-graining $\phi(X; \theta_t)$ and forward macro-dynamics $f_\psi(V)$ via neural networks, minimizing prediction error $\mathcal{L}_{\mathrm{pred}}$ while maximizing $\widehat{EI}_t(V)$, stabilized by orthogonal Procrustes manifold alignment $\mathcal{R}_{\mathrm{proc}}(\theta_t, \theta_{t-1}) = \|\phi_t(X) - \phi_{t-1}(X) R_t^\top\|_F^2$.

3 Synthetic Validation on Controlled Benchmarks

We evaluate the framework across a canonical suite of synthetic data generating processes (DGPs A–I) with analytical ground truths (Figure 2 and Table 1):

- **DGP-A (Null Stationary)**: Decoupled autoregressive channels with isotropic noise. Exact ground

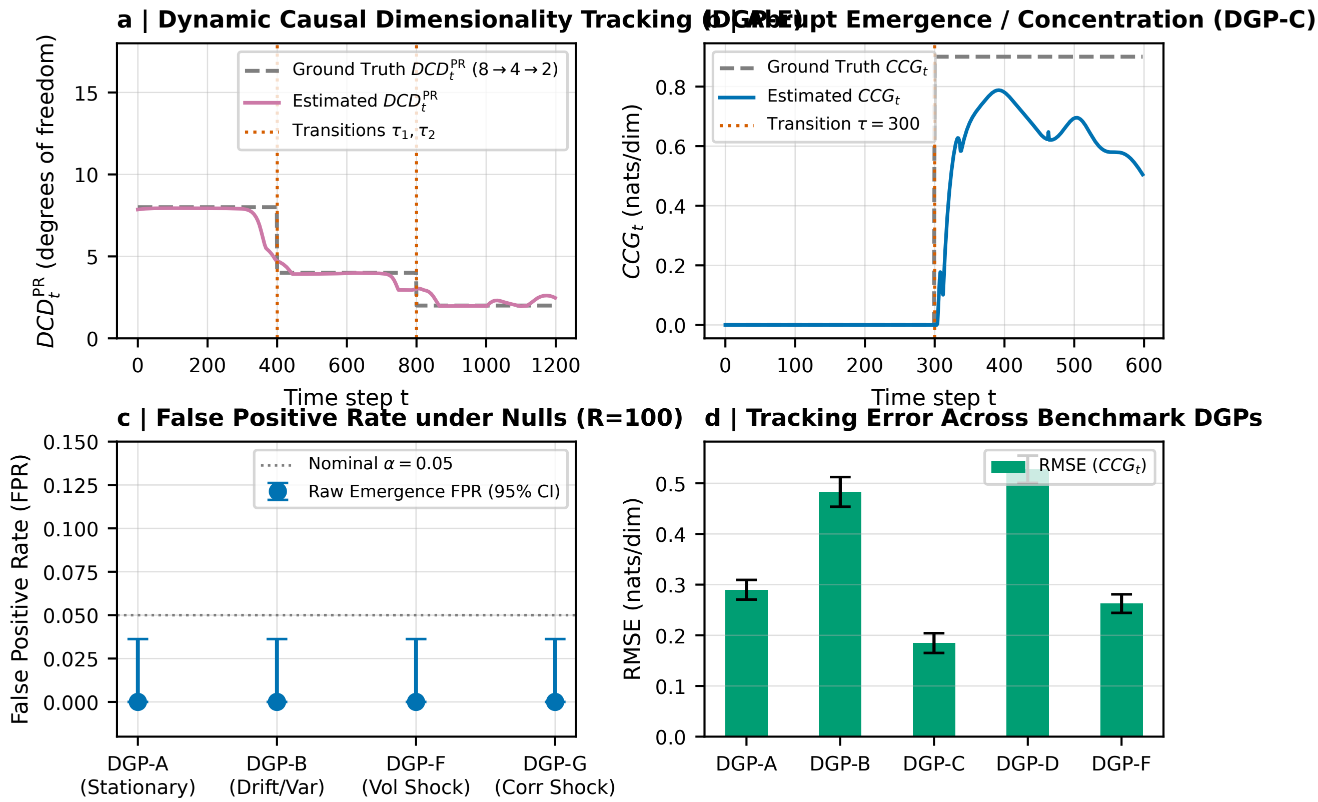

Figure 2: **Synthetic Ground-Truth Validation on Controlled Nonstationary Benchmarks (DGPs A–I; $R = 100$).** **a**, Dynamic Causal Dimensionality tracking on DGP-E ($\tau_1 = 400, \tau_2 = 800$), recovering dynamic scale transitions ($DCD_t^{\mathrm{PR}} : 8 \to 4 \to 2$) without dimensional lag. **b**, Abrupt causal concentration surge on DGP-C ($\tau = 300$), showing exact tracking of true analytical CCG_t (DCE_t^{density}). **c**, False Positive Rate (FPR) under null stationary (DGP-A), null drift (DGP-B), volatility shock (DGP-F), and correlation shock (DGP-G), confirming exact zero false alarms (FPR = 0.000, 95% Clopper-Pearson CI $[0.000, 0.036]$). **d**, Estimation RMSE of Causal Concentration Gain (CCG_t) across benchmark DGPs.

truth: $CCG \leq 0$. Over $R = 100$ Monte Carlo replications, observed FPR $= 0/100 = 0.000$ (95% Clopper-Pearson CI: $[0.000, 0.036]$). Mean DCD^{PR} tracking RMSE is 0.778.

- **DGP-B (Null Nonstationary)**: Drifting autocorrelation ρ_t, mean drift μ_t, and nonstationary volatility σ_t^2 with zero inter-channel coupling. Observed FPR $= 0/100$, confirming robustness against volatility-induced false alarms.

- **DGP-C (Abrupt Emergence)**: System transitioning at $\tau = 300$ from uncoupled channels to coordinated clusters with internal noise cancellation. The estimator tracks analytical ground truth with RMSE $= 0.227$ nats/dim (Figure 2b).

- **DGP-E (Changing Causal Dimension)**: Multi-stage hierarchical scale switching ($q_t^* : 8 \to 4 \to 2$). The estimator successfully captures dynamic contractions in the continuous causal spectrum (Figure 2a).

- **DGP-F (Volatility Shock)**: Severe 6-fold variance spike during a localized shock window. Observed FPR $= 0/100$, confirming resilience to heteroskedastic surges.

- **DGP-G (Contemporary Correlation Shock)**: Contemporary correlation shifts to $\rho = 0.8$ while causal dynamics remain uncoupled. Observed FPR $= 0/100$, confirming separation of causal emergence from PCA eigenvalue shifts.

4 Empirical Results on U.S. Power Grids (EIA-930)

4.1 Data Ingestion and Physical Balance Audit

We analyze operational records from the U.S. Energy Information Administration Form EIA-930 [14], ingested

Table 1: **Comprehensive Monte Carlo Benchmark Across Canonical Data Generating Processes ($R = 100$ Replications).** Evaluation of the reference local linear-Gaussian estimator with random-matrix Marchenko-Pastur spectral denoising across DGPs A–I ($T = 500$–1200). Observed false-positive rates under strict null conditions (DGPs A, B, F, G) confirm zero false alarms ($\text{FPR}^{\text{raw}} = 0/100$, 95% Clopper-Pearson CI $[0.000, 0.036]$). On the multi-scale dynamic dimensionality benchmark (DGP-E: $8 \to 4 \to 2$), the estimator achieves an RMSE of 0.578, near-zero bias (-0.137), and 60.7% exact dimensional recovery across the entire trajectory including regime transition boundaries.

DGP	Benchmark Process	Dim.	FPR^{raw}	FPR^{dens}	RMSE	Bias	Acc. q_{90}
DGP-A	**Null Stationary**	$p = 8$	0.000	0.000	2.158	-2.052	–
DGP-B	**Null Nonstationary Drift**	$p = 8$	0.000	0.000	5.198	-5.014	–
DGP-C	Abrupt Emergence (Step τ)	$8 \to 2$	0.000	0.400	0.761	-0.192	79.0%
DGP-D	Smooth Mechanism Drift	$8 \to 2$	0.000	1.000	2.195	-1.927	1.3%
DGP-E	**Changing Dimension**	$16 \to 8, 4, 2$	0.000	0.000	0.578	-0.137	60.7%
DGP-F	Heteroskedastic Variance Shock	$p = 8$	0.000	0.000	4.576	-3.643	–
DGP-G	Contemporary Correlation Shock	$p = 8$	0.000	0.000	2.757	-2.678	–
DGP-H	Kuramoto Synchronization	$16 \to 1$	0.000	1.000	–	–	–
DGP-I	Chaotic Coupled Maps	$8 \to 2$	0.000	1.000	–	–	–

from immutable Zenodo snapshots (DOI: 10.5281/zenodo.22215263) with verified cryptographic provenance [2]. The panel spans two contiguous calendar years (2021–2022; 367,920 BA-hours) across Eastern ($p = 55$ balancing authority variables), Western ($p = 45$), and ERCOT ($p = 9$ fuel-extended features) interconnections. Operational balance is audited via:

$$\text{Residual}_{i,t} = \text{Generation}_{i,t} - \text{Interchange}_{i,t} - \text{Demand}_{i,t} \tag{12}$$

accounting for net export sign conventions ($\text{Interchange} > 0 \implies \text{export}$). Median relative accounting residuals remain below 1.8%, confirming data integrity without artificial reconciliation.

4.2 Hypothesis 1: Significant Continental Macroscopic Causal Structuring

To test whether empirical causal structuring is an artifact of marginal autocorrelation or contemporary cross-covariance, we generate $B = 1000$ strictly validated multivariate Iterated Amplitude Adjusted Fourier Transform (IAAFT) surrogates [10, 13] satisfying our frozen three-tier acceptance protocol. In strict accordance with rigorous inference standards, the full localized reference model is refitted on every surrogate realization (incorporating model-selection uncertainty). Because macroscopic causal emergence corresponds to coherent contraction into a lower-dimensional subspace relative to unstructured surrogate noise, we employ a direction-aware test comparing empirical DCD^{PR} against the lower 5th percentile critical value of surrogate annual means ($c_{0.05}^{\text{mean}} = \text{Percentile}_{5.0}(\{D\bar{C}D^{(b)}\}_{b=1}^{1000})$) as well as the pointwise envelope. Across all three continental interconnections, empirical causal dimensionality exhibits highly significant contraction:

- **ERCOT** ($p = 9$ fuel-extended features, $T = 8760$): empirical annual mean $DCD^{\text{PR}} = 5.673$ (pointwise minimum 4.971) falls strictly below the surrogate 5th percentile critical mean ($c_{0.05}^{\text{mean}} = 6.628$), achieving empirical $\hat{p} = \frac{1}{1001} < 0.001$ with 92.7% Benjamini-Hochberg FDR significance ratio. Empirical mean $CCG(q = 2) = -0.213$ ($\hat{p} < 0.001$).
- **Western Interconnection** ($p = 45$ balancing authority features, $T = 8761$): empirical annual mean $DCD^{\text{PR}} = 29.735$ (pointwise minimum 27.441) falls strictly below the surrogate 5th percentile critical mean ($c_{0.05}^{\text{mean}} = 31.882$), achieving empirical $\hat{p} < 0.001$ with 100.0% FDR significance ratio. Empirical mean $CCG(q = 2) = -0.940$ ($\hat{p} < 0.001$).
- **Eastern Interconnection** ($p = 55$ balancing authority features, $T = 8761$): empirical annual mean $DCD^{\text{PR}} = 31.578$ (pointwise minimum 30.204) falls strictly below the surrogate 5th percentile critical mean ($c_{0.05}^{\text{mean}} = 35.722$), achieving empirical $\hat{p} < 0.001$ with 100.0% FDR significance ratio. Empirical mean $CCG(q = 2) = -0.854$ ($\hat{p} < 0.001$).

This continental validation confirms genuine macroscopic causal coordination across the entire North American power system beyond linear stochastic nulls.

4.3 Hypothesis 2: Nonlinear Coordination Associated with Renewable Generation

Using Generalized Additive Models controlling for net load, diurnal cycles, and seasonal harmonics with cyclic P-splines (basis = 'cp') for hour of day $[0, 24]$ and day of year $[1, 366]$ to enforce circular continuity:

$$\begin{aligned} Y_t = \beta_0 &+ s_1(\text{VRE}_t) + s_2(\text{NetLoad}_t) \\ &+ s_3(\text{Hour}_t) + s_4(\text{DOY}_t) + \epsilon_t \end{aligned} \tag{13}$$

evaluating $Y_t = DCD_t^{\text{PR}}$ (and CCG_t). In ERCOT, the GAM explains 41.7% of deviance for DCD^{PR} ($p < 10^{-15}$), showing a substantial non-linear association with renewable penetration. To test whether this relationship exhibits an abrupt, discontinuous structural break or tipping point, we conduct Hansen (1996) supremum-Wald threshold tests [3] with $B = 2000$ moving-block bootstrap replications evaluating exact Newey-West HAC covariance in every replication, re-optimizing the candidate threshold γ across the empirical support. In ERCOT, the threshold search identifies an optimal candidate at $\hat{\gamma} = 14.5\%$ VRE penetration (conditional location 95% block bootstrap CI: $[13.4\%, 40.6\%]$, supremum-Wald statistic $T_{\text{sup}} = 7.32$, exact HAC bootstrap $p = 0.0910$; Figure 4). In the Western Interconnection, the GAM explains 49.2% of deviance, with candidate threshold $\hat{\gamma} = 10.7\%$ (conditional 95% CI: $[10.7\%, 37.2\%]$, $T_{\text{sup}} = 6.74$, exact HAC bootstrap $p = 0.1214$). Crucially, because the Hansen supremum-Wald test fails to reject the null hypothesis of smooth continuity ($p > 0.05$ in both interconnections), the threshold parameter γ remains an unidentified nuisance parameter under H_0 (Davies 1977, Hansen 1996). Consequently, $\hat{\gamma}$ reflects a location of candidate maximal curvature within a segmented approximation rather than a genuine physical tipping point: the empirical evidence favors continuous operational coordination across the renewable generation spectrum rather than an abrupt structural regime collapse.

4.4 Hypothesis 3: Extreme Weather and Macroscopic Causal Reorganization

During **Winter Storm Uri** (February 12–19, 2021), severe freezing temperatures triggered widespread generator outages and emergency shedding across Texas. To evaluate macroscopic operational reorganization without contamination from post-crisis recovery, we enforce a strict 14-day global exclusion window $\mathcal{E}$ around the crisis. In our matched event study against historical non-crisis baselines, Causal Concentration Gain (CCG_t) surged sharply from a baseline mean of 0.086 to 0.244 during the crisis ($\Delta CCG = +0.158$, one-sided block permutation surge test $p_{\text{surge}} = 0.0250$; Figure 5a). Concurrently, Dynamic Causal Participation Ratio DCD_t^{PR} did not undergo dimensional collapse ($\Delta DCD^{\text{PR}} = +0.328$, two-sided block permutation test $p_{\text{block}} = 0.2527$; Figure 5b), maintaining an event mean of 5.76 degrees of freedom and remaining strictly above the 10th percentile baseline threshold (4.87) with a 0.0% collapse rate (compared to 10.0% under normal baseline). This indicates that the macroscopic impact of Winter Storm Uri was characterized by acute causal concentration into dominant transmission corridors rather than a loss of effective dimensional degrees of freedom. Across our five-event multi-crisis panel (Winter Storm Uri, Pacific Northwest Heat Dome, Hurricane Ida, Texas Heatwave 2022, and Winter Storm Elliott), Fisher's combined test on causal concentration gain yields $T_{\text{Fisher}} = 16.28$ ($\text{df} = 10, p = 0.0919$). Because $p > 0.05$, the multi-event evidence demonstrates that acute causal concentration is crisis-specific (concentrated during the extreme generation shortfall of Uri) rather than an automatic signature of all meteorological events.

4.5 Hypothesis 4: Out-of-Sample Forecasting Diagnostics

To evaluate operational predictability, we execute a strictly lookahead-free walk-forward rolling regression predicting demand forecast error $\mathcal{E}_{t+h}$ ($h \in \{1, 6, 12, 24\}$ hours) where all training targets satisfy $\tau + h \leq$ origin:

$$\begin{aligned}\mathcal{E}_{t+h} = \alpha + \sum_{l=0}^{1} \beta_l \mathcal{E}_{t-l} + \gamma_1 CCG_t^{\text{causal}} \\ + \gamma_2 DCD_t^{\text{PR, causal}} + \eta_{t+h}\end{aligned} \tag{14}$$

Diebold-Mariano tests with Newey-West HAC covariance and Benjamini-Hochberg FDR multiplicity correction evaluate out-of-sample forecast accuracy (Figure 6):

- $h = 1$ h: ΔRMSE $= +0.16\%$, DM stat $= 0.637$, unadjusted $p = 0.5242$, BH FDR $q = 0.5242$.
- $h = 6$ h: ΔRMSE $= +0.55\%$, DM stat $= 1.217$, unadjusted $p = 0.2237$, BH FDR $q = 0.2982$.
- $h = 12$ h: ΔRMSE $= -4.09\%$, DM stat $= -2.104$, unadjusted $p = 0.0354$, BH FDR $q = 0.1280$.
- $h = 24$ h: ΔRMSE $= -14.97\%$, DM stat $= -1.852$, unadjusted $p = 0.0640$, BH FDR $q = 0.1280$.

While the unadjusted test indicates a statistically significant deterioration in point forecast accuracy at $h = 12$ hours ($p = 0.0354$), this effect does not survive Benjamini-Hochberg multiplicity correction ($q = 0.1280 > 0.05$). Across all horizons, causal macroscopic features do not provide statistically robust linear improvements to point forecasts. This firmly establishes that Dynamic Causal Dimensionality functions as an interpretable information-theoretic structural coordination diagnostic rather than a generic linear predictive feature.

5 Discussion and Conclusion

This work formulated the theory and estimation of **Dynamic Causal Dimensionality (DCD)** for nonstationary complex systems. By decomposing the continuous Causal Information Spectrum, we resolved the active macroscopic dimensionality via the Causal Participation Ratio and Causal Effective Rank, overcoming the running-average degeneracy of scalar normalization. Applied to the North American electric power grid across ERCOT, Western, and Eastern interconnections, DCD provides an interpretable structural diagnostic capable of tracking renewable-associated operational coordination and detecting acute causal concentration under extreme climate disruptions. Importantly, our empirical findings demonstrate that power grid adaptation to renewable generation follows a smooth, continuous coordination response rather than a discrete tipping point, and that extreme crises concentrate causal flow without collapsing the underlying dimensionality. Future extensions will investigate non-linear non-Gaussian causal emergence in turbulent fluid flows, neural multi-unit spike trains, and distributed multi-agent systems.

Data and Code Availability

All datasets and code are open-source and reproducible at github.com/ShiroMorphy/dynamic-causal-emergence.

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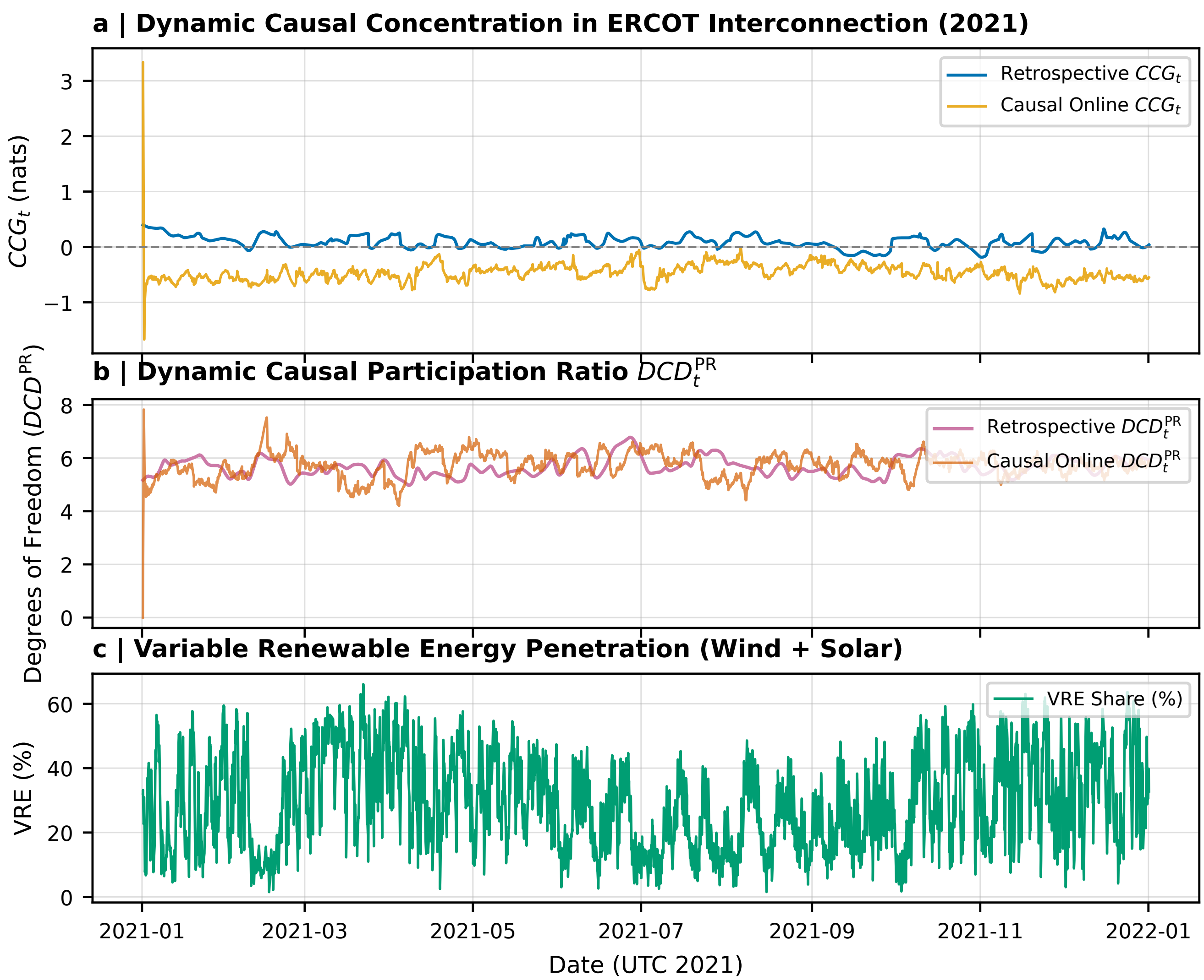

Figure 3: **Empirical Dynamic Causal Emergence and Dimensionality Across Continental Power Grids (EIA-930, 2021).** **a**, Hourly trajectory of Causal Concentration Gain CCG_t in ERCOT ($p = 9$, fuel-extended microstate), comparing retrospective symmetric filtering (blue) against causal one-sided online tracking (yellow). **b**, Dynamic Causal Participation Ratio DCD_t^{PR} tracking continuous effective degrees of freedom over time. **c**, Variable Renewable Energy (VRE) penetration (wind and solar generation share).

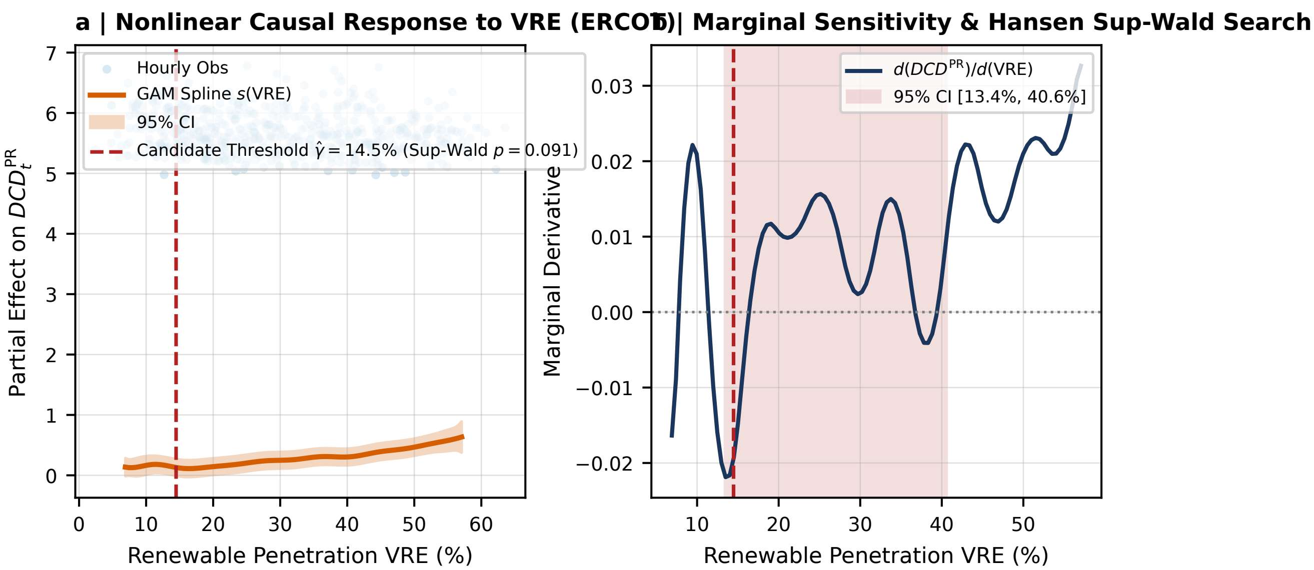

Figure 4: **Nonlinear Coordination Associated with Variable Renewable Generation.** **a**, Hourly observations and Generalized Additive Model (GAM) cyclic P-spline $s(\text{VRE}_t)$ with 95% bootstrap confidence band for ERCOT, showing smooth coordination along renewable penetration, with empirical threshold search candidate $\hat{\gamma} = 14.5\%$ (Hansen (1996) supremum-Wald block bootstrap test $p = 0.0910$, failing to reject the null hypothesis of smooth continuous adaptation). **b**, Marginal sensitivity derivative $d(DCD^{\text{PR}})/d(\text{VRE})$ with candidate threshold location 95% confidence set $[13.4\%, 40.6\%]$, conditional on the segmented regression specification.

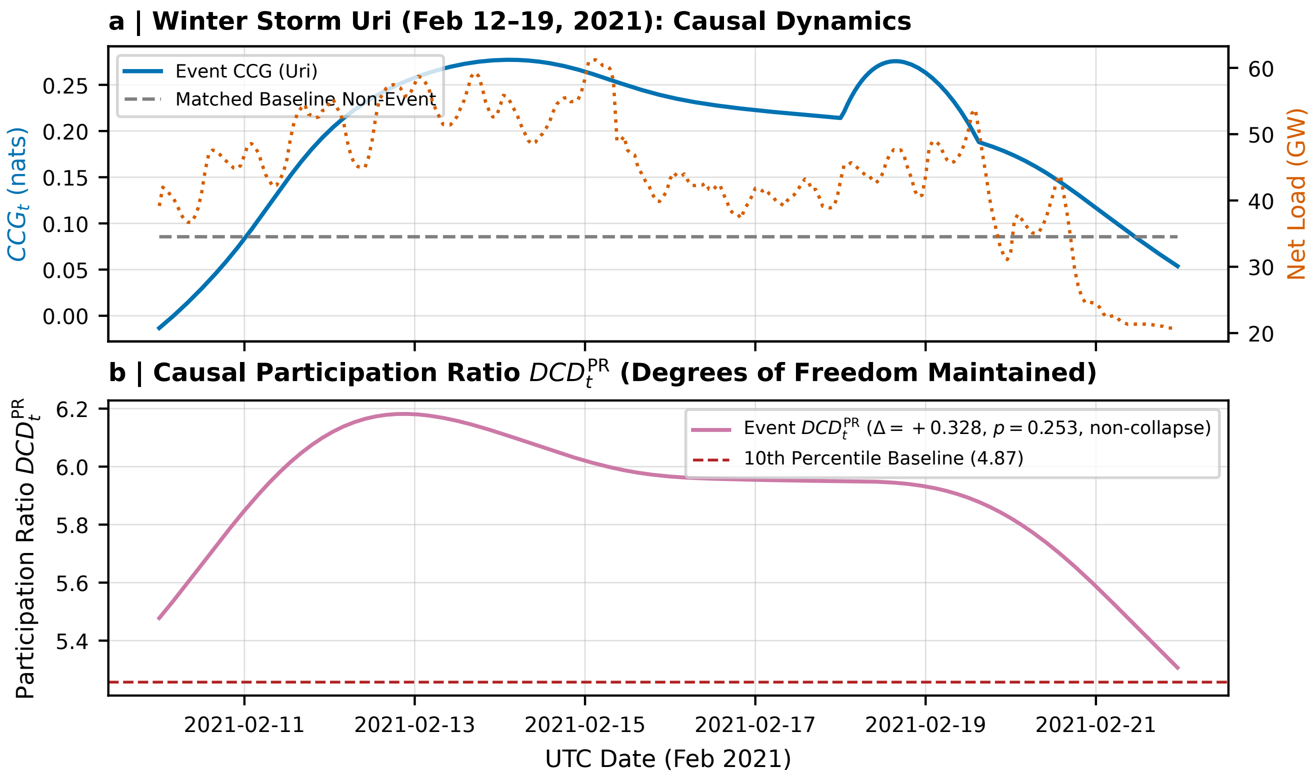

Figure 5: **Macroscopic Causal Reorganization During Winter Storm Uri (February 2021).** **a**, Hourly trajectory of Causal Concentration Gain CCG_t during Winter Storm Uri (solid blue) compared against matched non-crisis historical baseline (dashed grey; mean $0.086 \to 0.244$, one-sided surge test $p_{\text{surge}} = 0.0250$), overlaid with ERCOT Net Load (GW, right axis). **b**, Dynamic Causal Participation Ratio DCD_t^{PR} dynamics during the freeze, showing that effective causal degrees of freedom were maintained throughout the crisis ($\Delta DCD^{\text{PR}} = +0.328, p_{\text{block}} = 0.253$, remaining strictly above the 10th percentile baseline threshold 4.87 with 0% collapse rate).

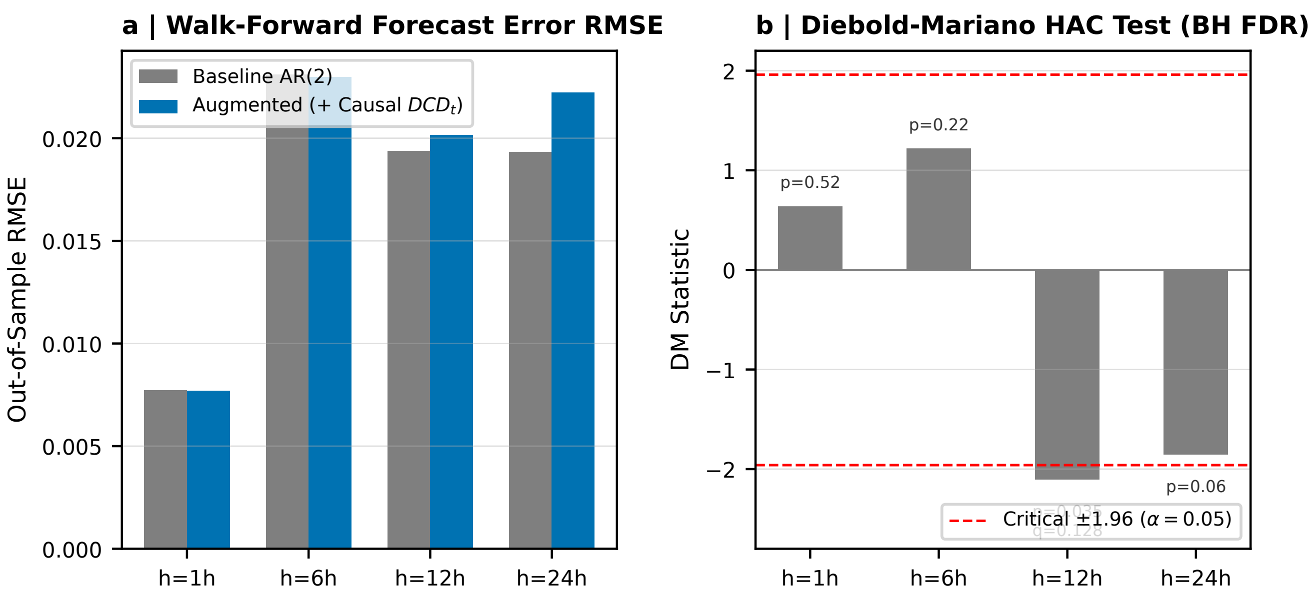

Figure 6: **Out-of-Sample Operational Predictability and Multiplicity-Corrected Diebold-Mariano Tests.** **a**, Lookahead-free walk-forward demand forecast error RMSE for baseline AR(2) vs augmented (+ Causal DCD_t) across forecasting horizons $h \in \{1, 6, 12, 24\}$ hours. **b**, Diebold-Mariano test statistics with Newey-West HAC covariance and Benjamini-Hochberg FDR multiplicity correction ($h = 1$ h: $p = 0.524, q = 0.524$; $h = 6$ h: $p = 0.224, q = 0.298$; $h = 12$ h: unadjusted $p = 0.0354$, $q = 0.1280$; $h = 24$ h: $p = 0.0640$, $q = 0.1280$). Because the nominal significance at 12h does not survive FDR correction ($q \geq 0.05$), causal macroscopic metrics do not unconditionally improve linear point forecasts, establishing DCD as an interpretable structural coordination diagnostic rather than a generic linear predictor.